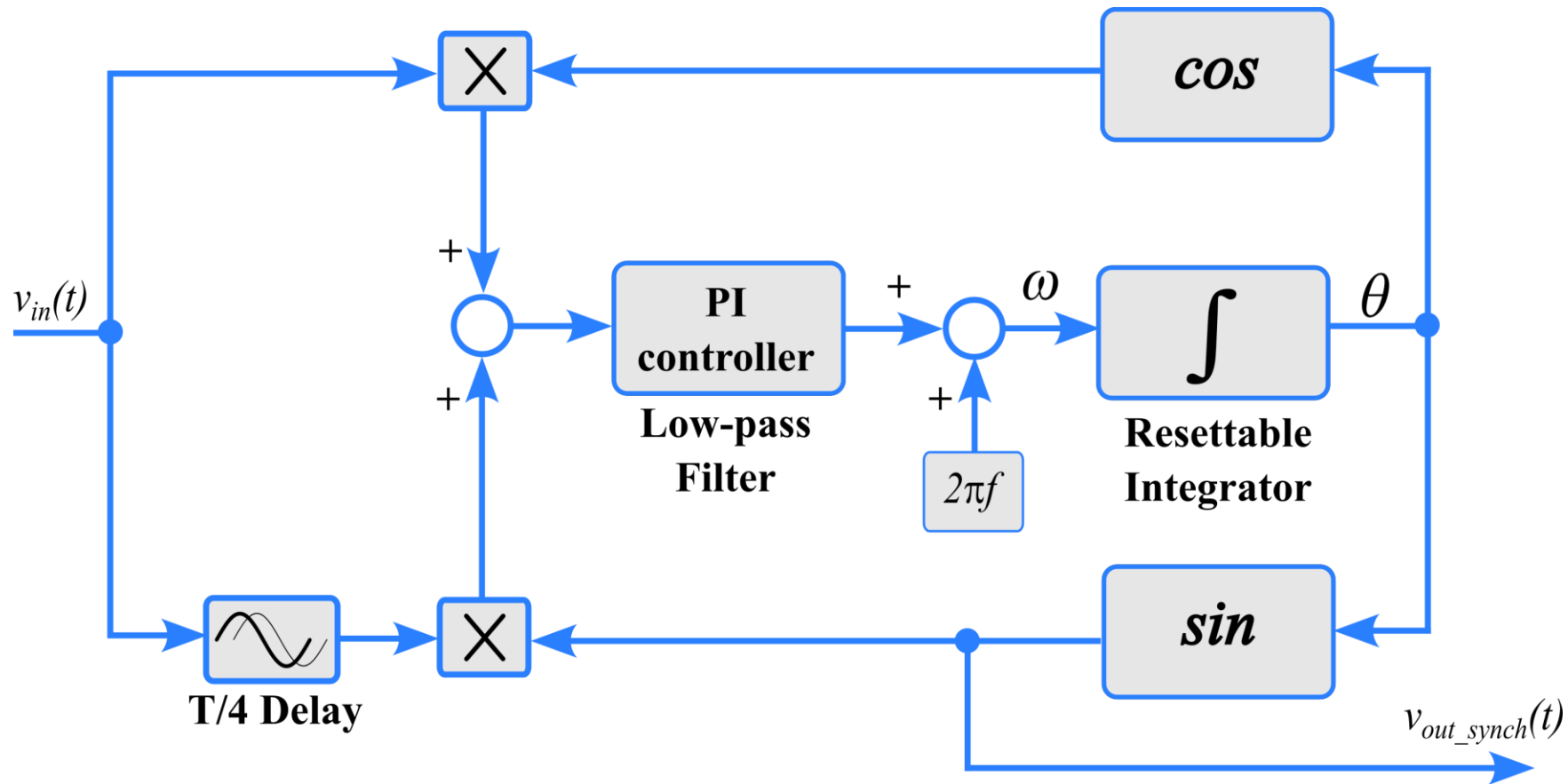


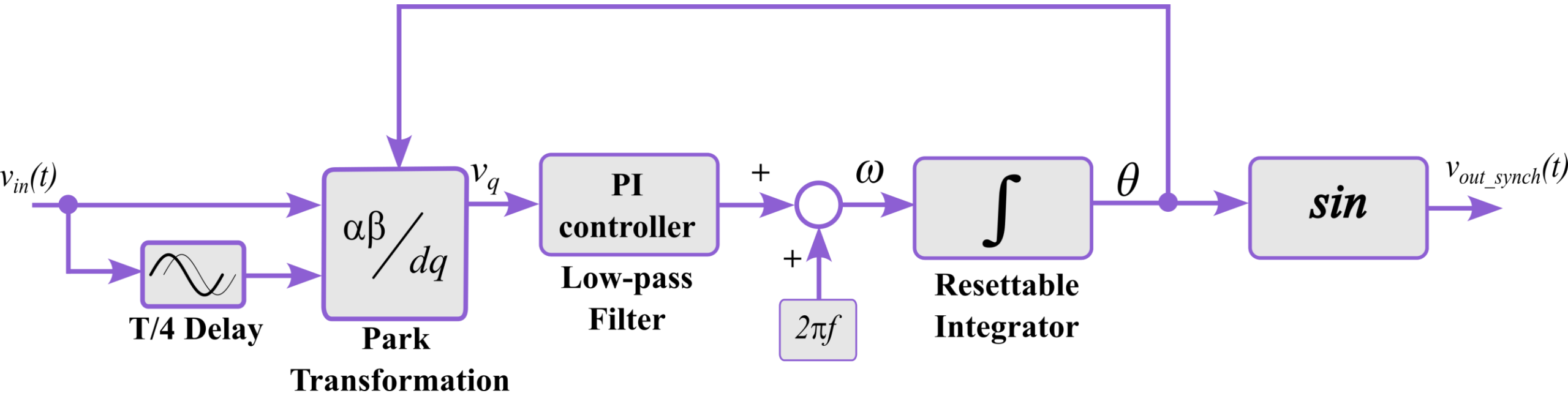
PLL melhorados

- Continuação do conteúdo apresentado no quadro

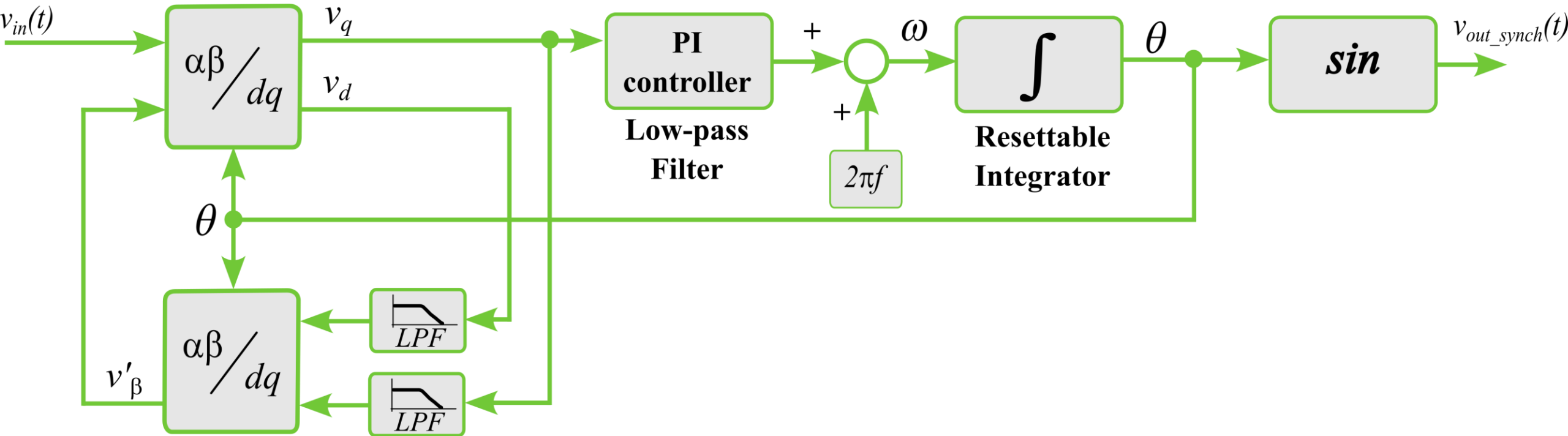
PLL baseado em sinais em quadratura ideal



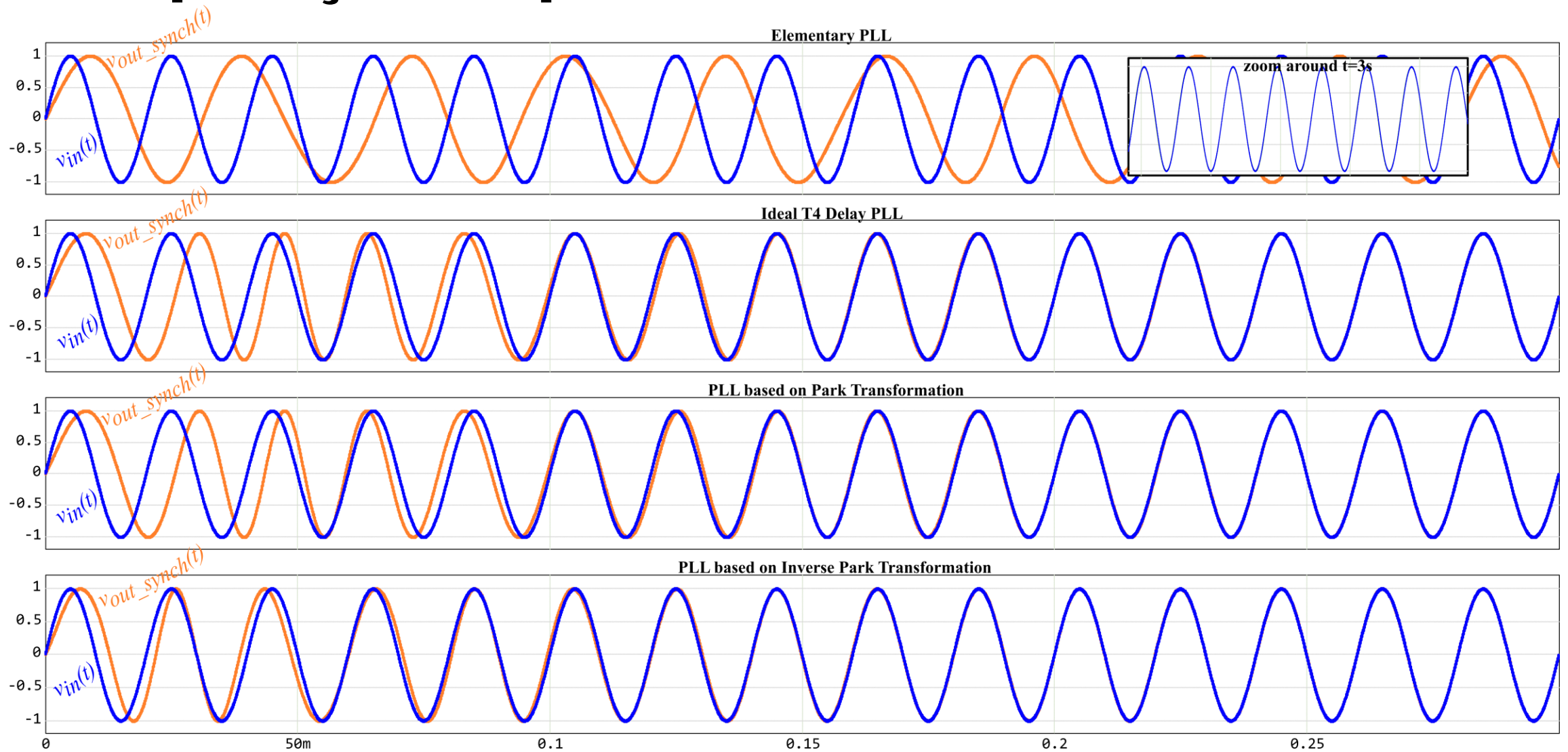
PLL baseado na Transformada Park



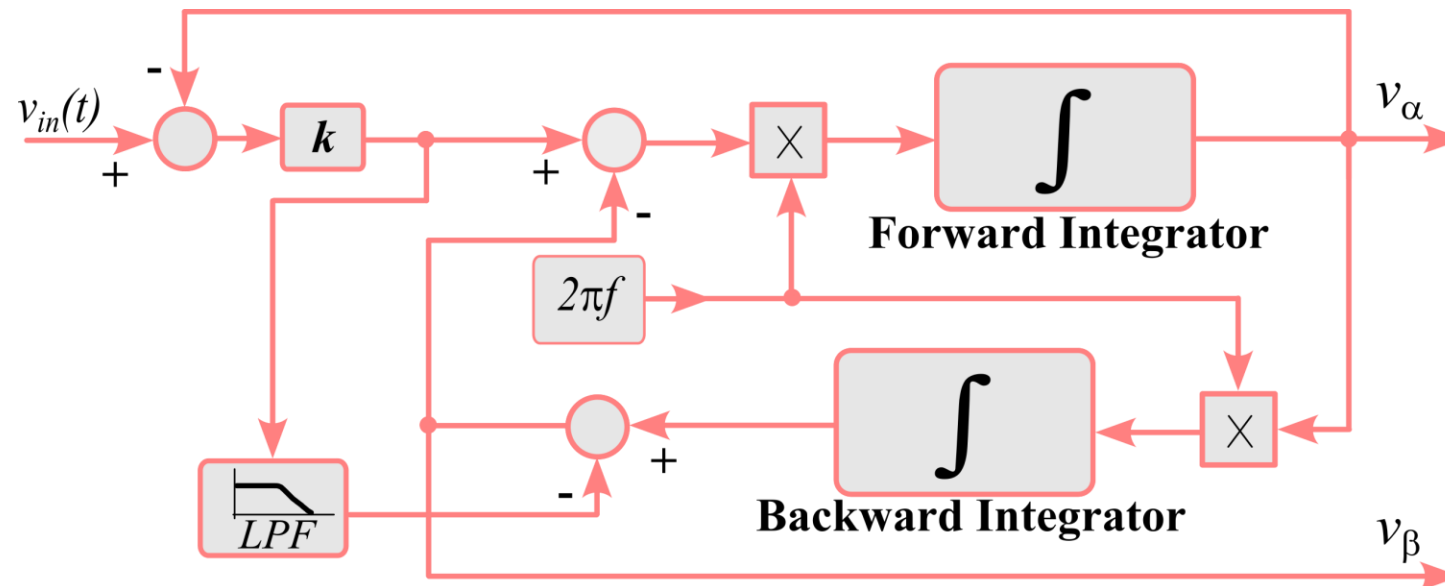
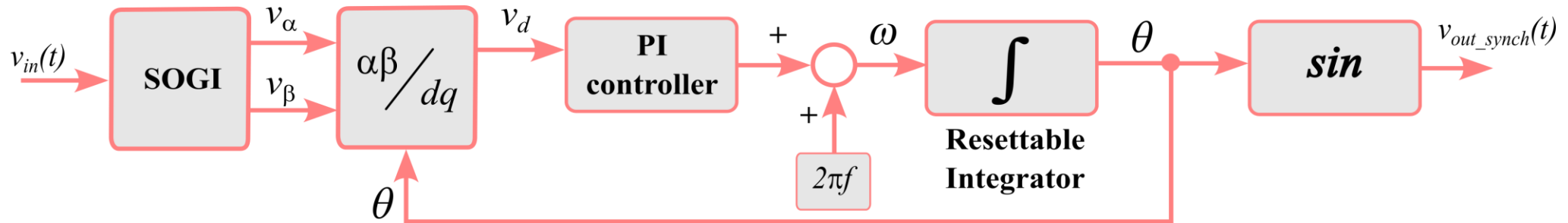
PLL baseado na transformada Park Inversa



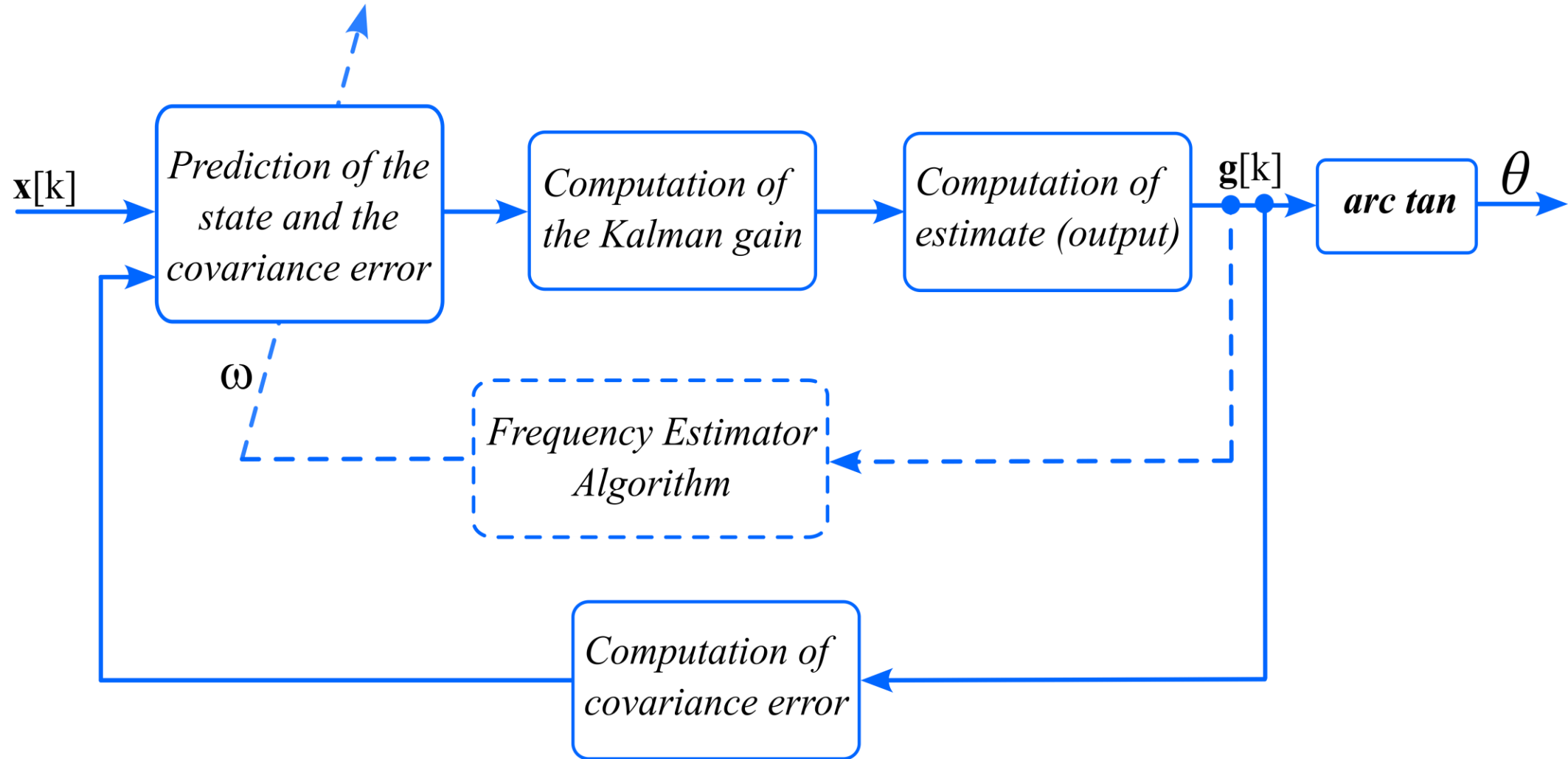
Comparação de performance



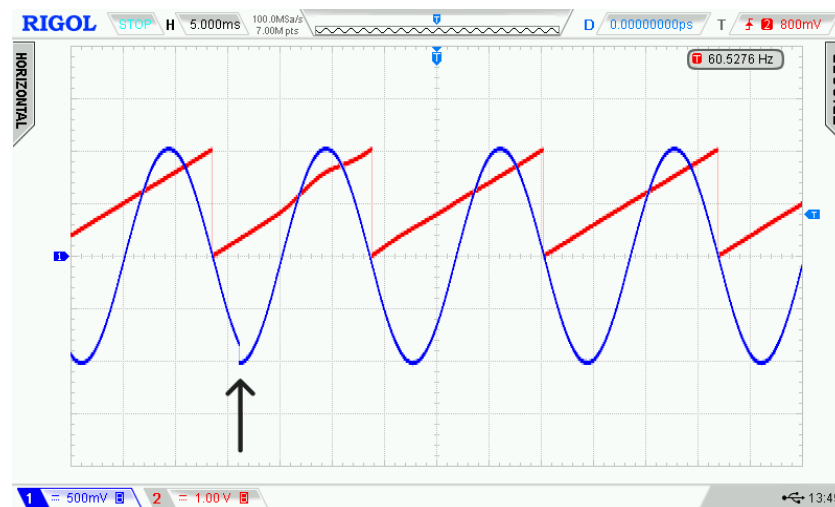
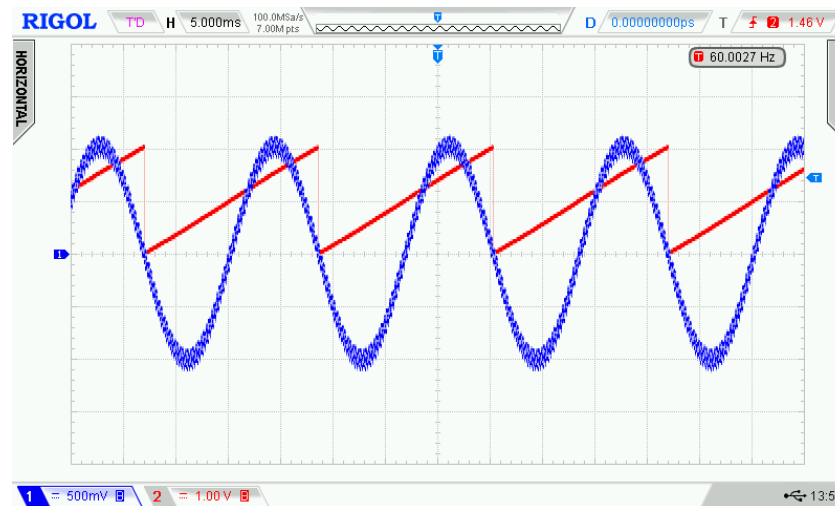
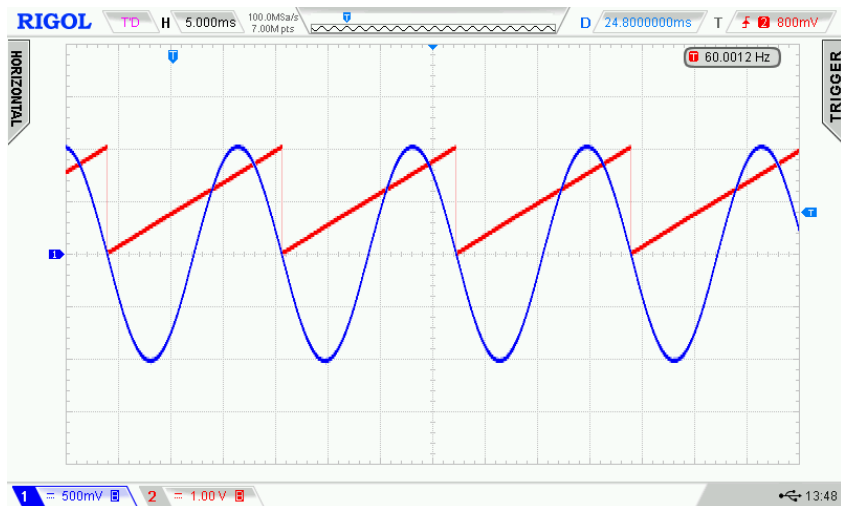
PLL baseado em SOGI



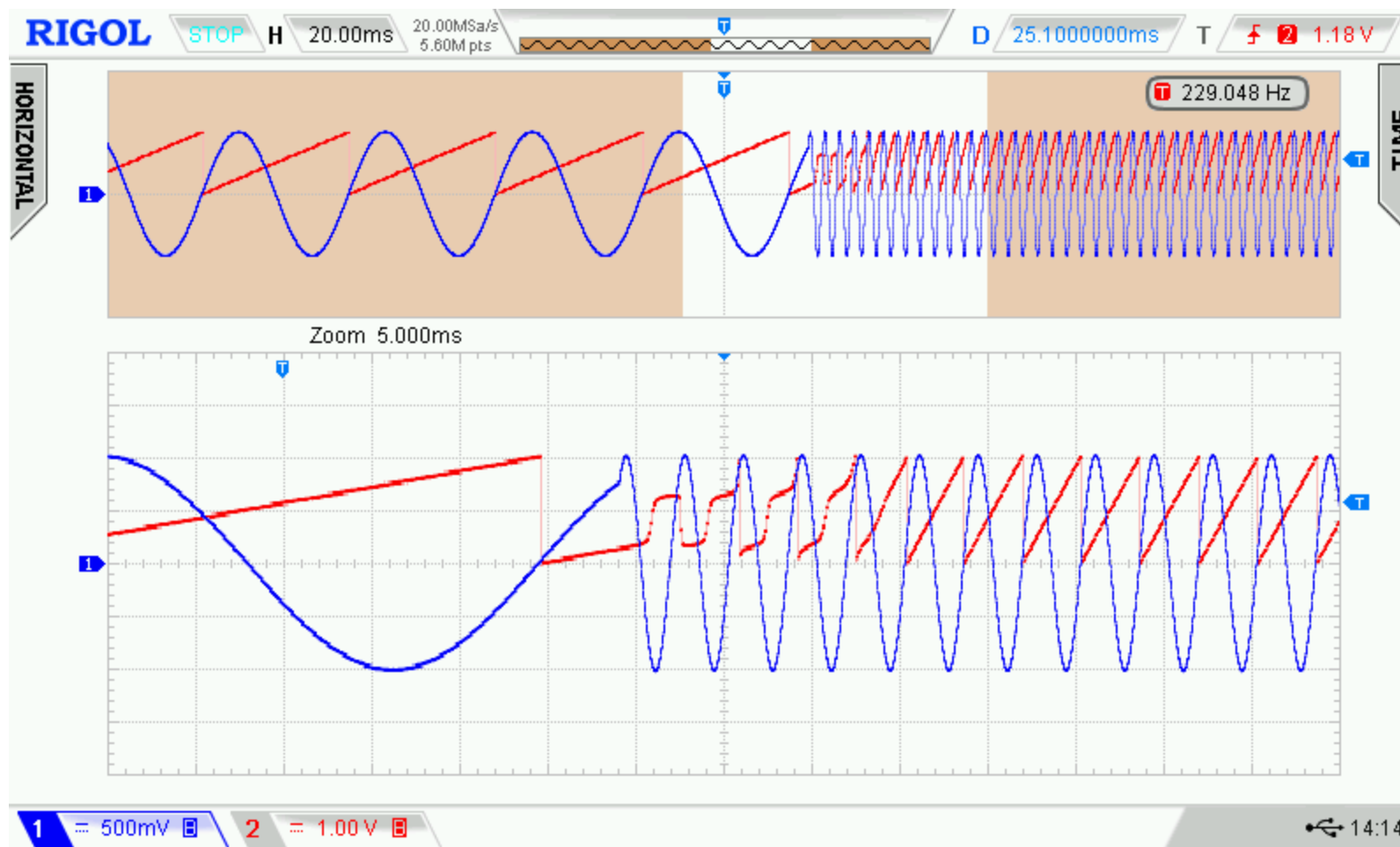
PLL baseado no filtro de Kalman



Testando os PLLs



PLL baseado em filtro de Kalman adaptativo em frequência



Referência.

[1] Tiago Davi Curi Busarello. "Phase-Locked Loops in Power Electronics: Principle and C-HIL-Based Verification". to be published in 2027 as a chapter in the book Model Based Engineering for CPS in Energy Technology, IET.